Enzymes

Mark scheme and worked solutions for 20 questions: 10 multiple choice, 10 structured.

1 · Part 1

Core · Enzymes · [1]

Answer: B

Explanation

A catalyst increases the rate of a chemical reaction and is not changed by the reaction. A is wrong because a catalyst is not used up; C is wrong because a catalyst speeds a reaction up, not down; D is wrong because a catalyst does not become part of the product.

2 · Part 1

Core · Enzymes · [1]

Answer: D

Explanation

Enzymes are proteins that act as biological catalysts in all metabolic reactions. A and C give the wrong type of molecule (and C wrongly limits them to digestion); B is wrong because, as catalysts, enzymes are not used up.

3 · Part 1

Core · Enzymes · [1]

Answer: A

Explanation

Enzymes speed up metabolic reactions so that they happen fast enough to keep the organism alive. B is wrong because enzymes do not supply energy; C is wrong because enzymes are not the only proteins in cells and this is not why they are needed; D is wrong because enzymes do not raise body temperature.

4 · Part 1

Core · Enzymes · [1]

Answer: C

Explanation

The shape of the active site is complementary to the shape of the substrate, so the substrate fits, and the reaction then forms products. A compares the wrong molecules; B is wrong because enzymes are catalysts and are not used up; D is wrong because the active site is not permanently changed by normal enzyme action.

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5 · Part 1 Core Enzymes [1]

Answer: A

Worked solution

Read down the rate column and find the largest value: 52 arbitrary units at 40 °C. Above 40 °C the rate falls sharply, which fits denaturation of the enzyme at high temperature.

6 · Part 1 Extended Enzymes [1]

Answer: D

Explanation

pH 8 is far from the optimum pH of 2, so the enzyme's shape, including the active site, changes (denaturation) and the substrate is no longer complementary, so no enzyme-substrate complexes form. A confuses pH with temperature; B is wrong because enzymes are not used up; C is wrong because a closer fit would increase activity, not stop it.

7 · Part 1 Extended Enzymes [1]

Answer: B

Explanation

The substrate binds to the active site to form an enzyme-substrate complex; the products are then released and the enzyme is unchanged. A and D reverse or mix up substrate and product; C is wrong because the enzyme is not denatured by catalysing a reaction.

8 · Part 1 Extended Enzymes [1]

Answer: C

Explanation

Enzymes are specific: only a substrate with a shape complementary to the active site can fit and form an enzyme-substrate complex. Starch does not have a complementary shape. A is wrong because molecules in a mixture collide randomly, so starch and lipase do collide, but starch does not fit the active site; B is wrong because starch does not denature lipase; D is wrong because enzymes are not used up.

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9 · Part 1 Extended Enzymes [1]

Answer: A

Explanation

Below the optimum, a rise in temperature gives the molecules more kinetic energy, so they move faster and collide more often, forming more enzyme-substrate complexes per second. B is a common misconception: heating does not improve the fit; C is wrong because denaturation reduces activity; D is wrong because temperature does not make more enzyme molecules in the reaction mixture.

10 · Part 1 Extended Enzymes [1]

Answer: C

Worked solution

Compare the two samples. Both were tested at 37 °C, so any difference must come from the earlier treatment. High temperature changes the shape of the enzyme permanently (denaturation), so sample 1 cannot form enzyme-substrate complexes even after cooling. Low temperature only slows molecules down; when sample 2 was warmed again its active site was still the correct shape.

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11 · Part 1 Core Enzymes [5]

Mark scheme

(a) **[2]**

- (a substance that) increases the rate of a (chemical) reaction (1)
- is not changed / not used up by the reaction (1)

ALLOW speeds up a reaction

REJECT starts a reaction that could not happen

(b) **[1]**

- protein (1)

(c) **[2]**

- (enzymes) speed up / catalyse (all) metabolic reactions (1)
- without enzymes, reactions would be too slow to sustain life / keep the organism alive (1)

ALLOW reactions happen fast enough to keep cells alive

Worked solution

(a) A catalyst is a substance that increases the rate of a chemical reaction and is not changed by the reaction, so the same catalyst can be used again and again.

(b) Enzymes are proteins.

(c) Enzymes are biological catalysts involved in all metabolic reactions. Without them, these reactions would happen far too slowly for the organism to stay alive. Enzymes make the reactions fast enough to sustain life.

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12 · Part 1 Core Enzymes [5]

Mark scheme

(a) **[2]**

- substrate: lactose (1)
- products: glucose and galactose (both needed) (1)

(b) **[3]**

- lactose / substrate fits into the active site (of lactase) (1)
- the shape of the active site is complementary to (the shape of) lactose (1)
- products (glucose and galactose) are formed and leave the active site / the enzyme is unchanged and can be used again (1)

ALLOW enzyme-substrate complex forms, for the first mark

REJECT active site is the same shape as lactose

Worked solution

(a) The substrate is the molecule the enzyme acts on, lactose. The products are the molecules made, glucose and galactose.

(b) Lactase has an active site with a shape complementary to lactose. A lactose molecule fits into the active site and is broken down into glucose and galactose. These products leave the active site, and the lactase is unchanged, so it can break down another lactose molecule. Remember: "complementary" means the shapes fit together, not that they are identical.

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13 · Part 1 Core Enzymes [7]

Mark scheme

(a) **[1]**

- 40 °C (1)

(b) **[3]**

- from 10 °C to 40 °C, activity increases / time taken decreases (1)
- above 40 °C (to 70 °C), activity decreases / time taken increases (1)
- at 70 °C there is no activity / starch is not broken down (1)

ALLOW one correct use of data from the table (temperature with time) in place of any one of the points above

IGNORE references to optimum temperature on their own

(c) **[2]**

- the enzyme / amylase is denatured (1)
- the shape of the active site has changed, so starch no longer fits / is no longer complementary (1)

REJECT enzyme is killed / enzyme is dead

(d) **[1]**

- any one from: volume / concentration of starch solution; volume / concentration of amylase solution; pH (1)

IGNORE amount, unless qualified

REJECT temperature

Worked solution

(a) The shortest time means the fastest reaction. The shortest time, 90 s, is at 40 °C.

(b) Remember that a shorter time means higher activity. Between 10 °C and 40 °C the time falls from 600 s to 90 s, so activity increases. Above 40 °C the time rises again (300 s at 55 °C), so activity decreases. At 70 °C starch is still present after 900 s, so the amylase is not working.

(c) At 70 °C the amylase has been denatured. Its shape, including the active site, has changed, so starch molecules no longer fit and cannot be broken down.

(d) Only temperature should change. Everything else that could affect the rate, such as the volume and concentration of each solution and the pH, must be kept the same.

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14 · Part 1 Core Enzymes [5]

Mark scheme

(a) **[1]**

- pH 2 (1)

(b) **[1]**

- it took longer (to become clear) at pH 3 / 8 minutes compared with 6 minutes (1)

(c) **[2]**

- the protease is denatured (at pH 9) (1)
- the shape of the active site has changed, so the protein / egg white no longer fits / is no longer complementary (1)

ALLOW pH 9 is too far from the optimum pH, for the first mark only if denaturation is not mentioned

REJECT enzyme is killed

(d) **[1]**

- to make the results more reliable / to identify anomalous results / to calculate a mean (1)

REJECT to make the results more accurate

Worked solution

(a) The shortest time (6 min) is at pH 2, so pH 2 is the optimum of the values tested.

(b) A slower reaction takes longer. At pH 3 the suspension took 8 minutes to clear, but only 6 minutes at pH 2.

(c) At pH 9 the pH is far from the optimum. The protease is denatured: its shape, including the active site, changes so that the protein substrate no longer fits. No protein is broken down, so the suspension stays cloudy.

(d) Repeating the investigation shows whether results are consistent, lets anomalous results be spotted, and allows a mean to be calculated, so the results are more reliable.

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15 · Part 1 Core Enzymes [6]

Mark scheme

(a) **[2]**

- fish enzyme: 10 °C (1)
- human enzyme: 40 °C (1)

(b) **[2]**

- the fish enzyme has higher activity than the human enzyme at every temperature from 0 °C to 20 °C (1)
- the activity of the human enzyme increases (from 5 to 30) / the activity of the fish enzyme increases then decreases / peaks at 10 °C (1)

ALLOW comparative use of data at one temperature, e.g. at 0 °C the fish enzyme is 60 and the human enzyme is 5, for one mark

(c) **[2]**

- the fish's body is at about 0 °C / low temperature (1)
- (enzyme activity high so) metabolic reactions are fast enough to keep the fish alive / sustain life (1)

ALLOW named reaction, e.g. respiration, happens fast enough

IGNORE the fish would freeze

Worked solution

(a) Read the highest activity in each column: 85 at 10 °C for the fish enzyme and 90 at 40 °C for the human enzyme.

(b) A comparison needs both enzymes in the same statement. From 0 °C to 20 °C the fish enzyme is always more active than the human enzyme. The human enzyme's activity rises steadily over this range, while the fish enzyme's activity rises to a peak at 10 °C and then falls.

(c) The fish's body is as cold as the sea, close to 0 °C. All its metabolic reactions are catalysed by enzymes. An enzyme that works well at low temperature keeps these reactions fast enough to sustain life.

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16 · Part 1 Extended Enzymes [6]

Mark scheme

(a) **[4]**

- maltose / substrate collides with / enters the active site (1)
- the active site has a shape complementary to maltose (1)
- an enzyme-substrate complex forms (1)
- maltose is broken down into glucose / the product, which leaves the active site (1)
- the enzyme is unchanged / can be used again (1)
- max 4

REJECT active site is the same shape as maltose

(b) **[2]**

- sucrose has a different shape / is not complementary to the active site of maltase (1)
- so sucrose cannot fit into the active site / no enzyme-substrate complex forms (1)

ALLOW enzymes are specific, for the first mark only if no reference to shape

Worked solution

(a) Maltose molecules move around and collide with maltase. The active site of maltase has a shape complementary to maltose, so maltose fits into it and an enzyme-substrate complex forms. The maltose is broken down into two glucose molecules (the product), which leave the active site. Maltase is unchanged and can catalyse the breakdown of more maltose.

(b) Enzymes are specific. Although sucrose is also made of two sugar units, its shape is not complementary to maltase's active site, so it cannot fit and no enzyme-substrate complex can form.

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17 · Part 1 Extended Enzymes [7]

Mark scheme

(a) **[2]**

- (21.0 − 6.0) ÷ 6.0 × 100 / 15.0 ÷ 6.0 × 100 (1)
- 250 (%) (1)

ALLOW 250 % with no working for both marks

(b) **[3]**

- (as temperature increases) enzyme and substrate molecules gain (more) kinetic energy (1)
- molecules move faster / collide more frequently (1)
- more effective collisions / more enzyme-substrate complexes formed per unit time (1)

REJECT enzyme gets more active sites / active site fits better

(c) **[2]**

- use smaller temperature intervals / more temperatures, e.g. every 2 °C or every 5 °C (1)
- between 35 °C and 55 °C (1)

ALLOW between 35 °C and 55 °C with any narrower range around 45 °C, e.g. 40 °C to 50 °C, for the second mark

Worked solution

(a) Increase = 21.0 − 6.0 = 15.0 cm^3. Percentage increase = increase ÷ original value × 100 = 15.0 ÷ 6.0 × 100 = 250 %. Check: 6.0 cm^3 increased by 250 % gives 6.0 + 15.0 = 21.0 cm^3.

(b) As temperature rises towards the optimum, the enzyme and substrate molecules gain kinetic energy. They move faster and collide more often, so there are more effective collisions and more enzyme-substrate complexes form each second. More hydrogen peroxide is broken down, so more oxygen is collected in 2 minutes.

(c) The largest volume is at 45 °C, but the true optimum could be anywhere between 35 °C and 55 °C. Repeating the investigation at small temperature steps in this range, for example every 2 °C, would find the optimum more precisely.

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18 · Part 1 Extended Enzymes [5]

Mark scheme

(a) **[1]**

- pH 4 and pH 6 (both needed) (1)

(b) **[3]**

- pH 8 is far from the optimum pH of enzyme X / enzyme X is denatured (1)
- the shape of the active site changes (1)
- the substrate no longer fits / is no longer complementary, so no enzyme-substrate complexes form (1)

(c) **[1]**

- enzyme Y was denatured at pH 2 / the change (to the active site) is permanent / not reversed (1)

Worked solution

(a) Both enzymes must have an activity above 0. Enzyme Y is 0 at pH 2 and enzyme X is 0 at pH 8 and pH 10, so only pH 4 and pH 6 have activity for both.

(b) Enzyme X works best in acid conditions. At pH 8 it is denatured: its shape, including the active site, changes. The substrate is no longer complementary to the active site, so it cannot fit and no enzyme-substrate complexes form.

(c) Even when enzyme Y was moved to its optimum pH, it still did not work. This shows that pH 2 had denatured enzyme Y, and the change in shape of its active site was permanent.

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19 · Part 1 Extended Enzymes [6]

Mark scheme

(a) **[1]**

- (enzymes are) specific / each enzyme acts on only one substrate (1)

(b) **[2]**

- starch would not be broken down / would be broken down more slowly (1)
- the active site is no longer complementary to starch, so starch cannot fit / fewer or no enzyme-substrate complexes form (1)

(c) **[3]**

- after all the starch has been broken down, add more starch to the same mixture (1)
- test samples (of the mixture) with iodine solution at intervals (1)
- the new starch is also broken down / the iodine solution stays orange-brown again, showing the enzyme still works (1)

ALLOW suitable alternative test for starch

ALLOW first test the mixture to show that no starch is left, for the second mark

Worked solution

(a) Each enzyme broke down only one of the three substrates, so enzymes are specific.

(b) Enzyme P breaks down starch because its active site is complementary to starch. If the shape of the active site changes, starch no longer fits, so few or no enzyme-substrate complexes form and starch is broken down slowly or not at all.

(c) A catalyst is not changed by the reaction. Wait until a test with iodine solution shows no starch is left, then add more starch to the same tube without adding more enzyme. Keep testing with iodine solution. If the new starch also disappears, the original enzyme P must still be there and working, so it was not used up.

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20 · Part 1 Extended Enzymes [7]

Mark scheme

(a) **[2]**

- 20 ÷ 5 = 4 (1)
- mg per min / mg/min (1)

ALLOW 4.0

(b) **[2]**

- product is formed faster at 60 °C during the first minute(s) / more product at 60 °C at 1 and 2 minutes (1)
- at 30 °C product forms at a steady rate throughout / more product in total at 30 °C (20 mg compared with 14 mg) (1)

ALLOW product formation at 60 °C stops after 3 minutes

(c) **[3]**

- (at 60 °C) the enzyme is denatured (over time) (1)
- the shape of the active site changes (1)
- the substrate no longer fits / is no longer complementary, so no enzyme-substrate complexes form (1)

REJECT substrate used up / runs out (the question states substrate is still present)

Worked solution

(a) Mean rate = total product ÷ time = 20 mg ÷ 5 min = 4 mg per min. Check: the mass rises by 4 mg every minute at 30 °C.

(b) At 60 °C, product forms faster at first (9 mg after 1 minute compared with 4 mg at 30 °C), because the molecules have more kinetic energy. After 3 minutes no more product forms at 60 °C. At 30 °C product forms at a steady rate, so after 5 minutes there is more product at 30 °C (20 mg) than at 60 °C (14 mg).

(c) Substrate is still present, so it is not the substrate running out. At 60 °C the enzyme molecules are gradually denatured: their shape, including the active site, changes. The substrate no longer fits the active site, no enzyme-substrate complexes form, and the reaction stops.

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